Exploring the Western Outskirts of the Small Magellanic Cloud: Tracing Foreground and SMC Stars with UVIT

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Abstract

The Ultra-Violet Imaging Telescope (UVIT), combined with the precise astrometry and optical photometry from *Gaia*, enables efficient separation and characterization of hot and evolved stellar populations in nearby galaxies that are difficult to distinguish using optical data alone. We apply this approach to a $28'$ region in the western outskirts of the Small Magellanic Cloud (SMC) centered around the cluster Kron 3. The FUV-*Gaia* vector point diagrams (VPDs) shows the SMC along with diffuse foreground population, whereas the NUV-*Gaia* VPDs additionally reveals a foreground clump. The foreground clump is identified as the members of the Galactic globular cluster 47 Tuc (NGC 104), which in color magnitude diagrams (CMDs) overlaps the SMC red clump region while following an evolutionary track characteristic of an old stellar population.

Introduction

SMC is a nearby low-metallicity dwarf galaxy at ~ 62kpc whose structure has been shaped by interactions with the Large Magellanic Cloud and the Milky Way. Its western halo move outward from the SMC, tracing active tidal interaction and mass loss, making it an ideal laboratory for studying tidal stripping and galaxy interaction dynamics, though severe foreground contamination from Milky Way stars and the globular cluster 47 Tuc complicates such analyses.

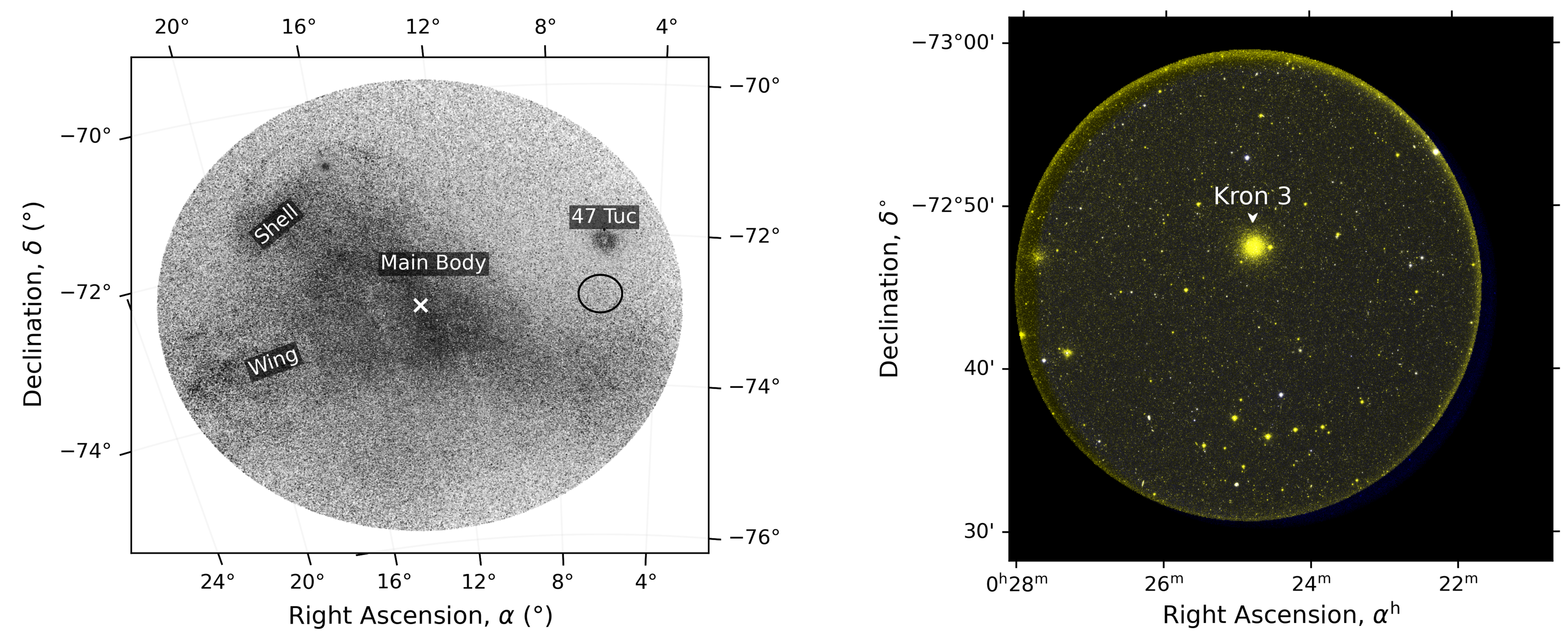

Figure 1. (a) Our studied UVIT region (black circle) over Gaia background. (b) False-colour composite of the studied UVIT field, the SMC globular cluster Kron 3 is visible near the centre.

Study methodology

- UVIT observations were obtained in the FUV (F148W) and NUV (N242W) filters, centred at RA = $6.191°$ and Dec = $-72.795°$.
- Level 1 UVIT data were processed using the CCDLAB pipeline.
- Stellar photometry was performed on the UVIT images following standard procedures.

Data Summary: FUV and NUV source counts from UVIT photometry to Probable SMC member .

Stage of Analysis	FUV	NUV
After Photometry	797	9187
After quality cuts[1]	505	8888
UVIT–*Gaia* matched[2]	191	2218
Probable SMC members[3]	103	1298

Notes.
1. Magnitude uncertainty ≤ 0.3 mag and saturation correction applied.
2. Matching radius of $1''$.
3. Probable SMC stars selected based on the Jiménez Gaia-based probability catalog, adopting $P_{cut} > 0.31$.

Results and discussion

- **FUV VPD:** Shows the SMC population along with a kinematically diffuse foreground population.
- **NUV VPD:** Reveals an additional distinct foreground clump not seen in the FUV VPD.

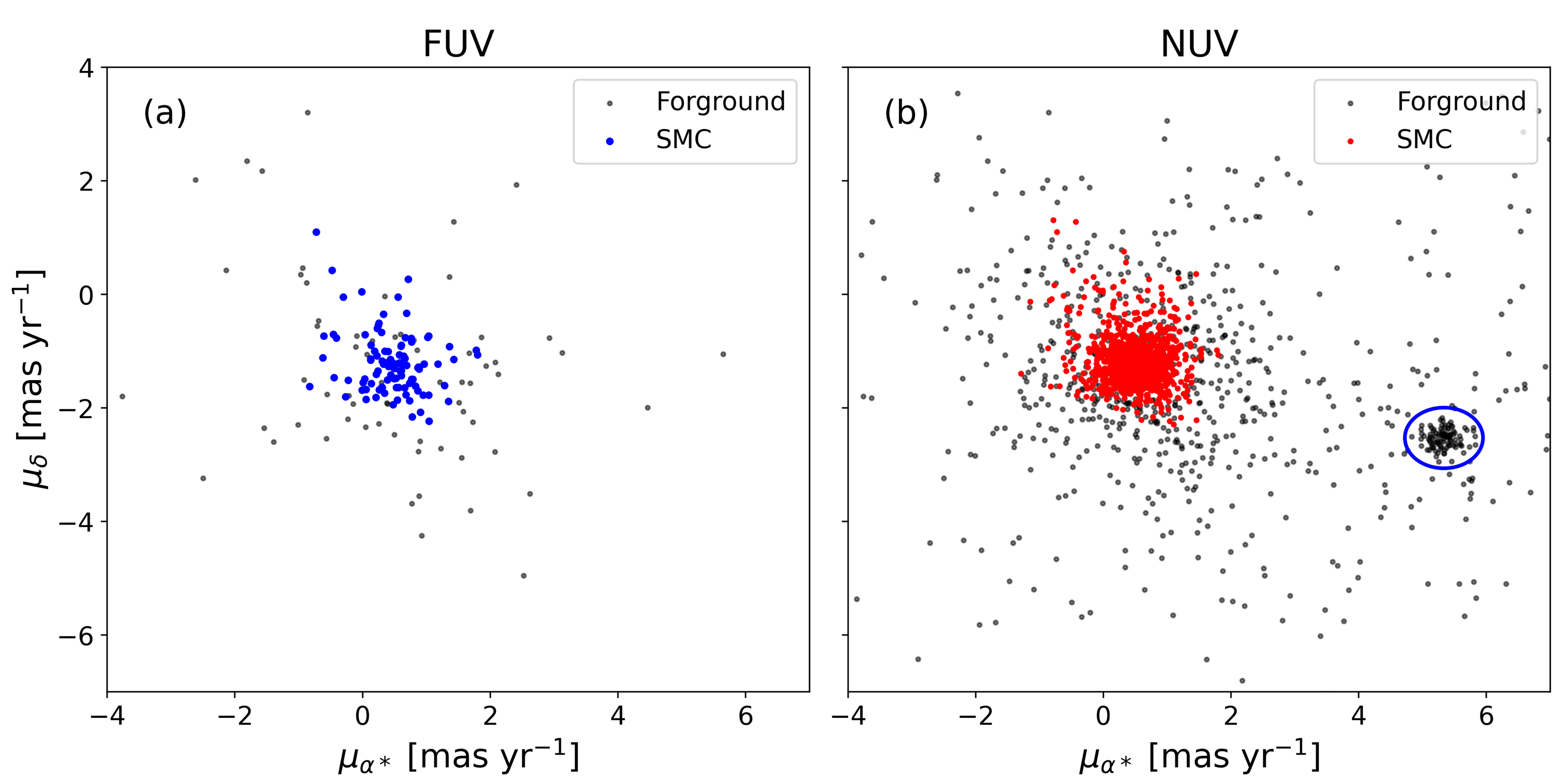

Figure 2. **VPDs:** (a) **FUV-SMC** in blue. (b) **NUV-SMC** in red and a secondary foreground clump enclosed by a blue ellipse. Grey corresponding to the Foreground population in both.

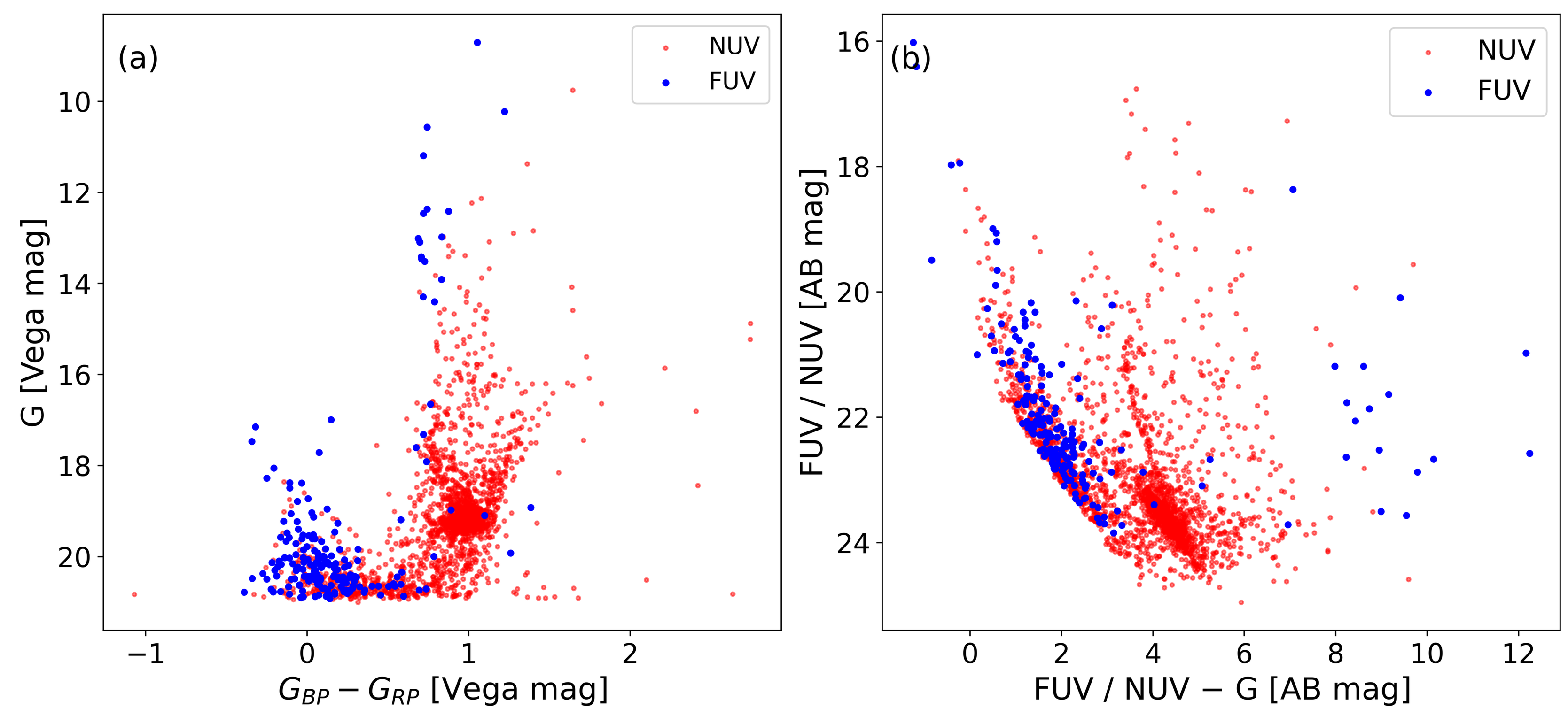

Figure 3. **CMDs:** (a) **Optical** and (b) **FUV/NUV–optical** of detected stars. In panel (b), a sequence of stars parallel to the NUV main sequence is visible, possibly indicating a distinct stellar population.

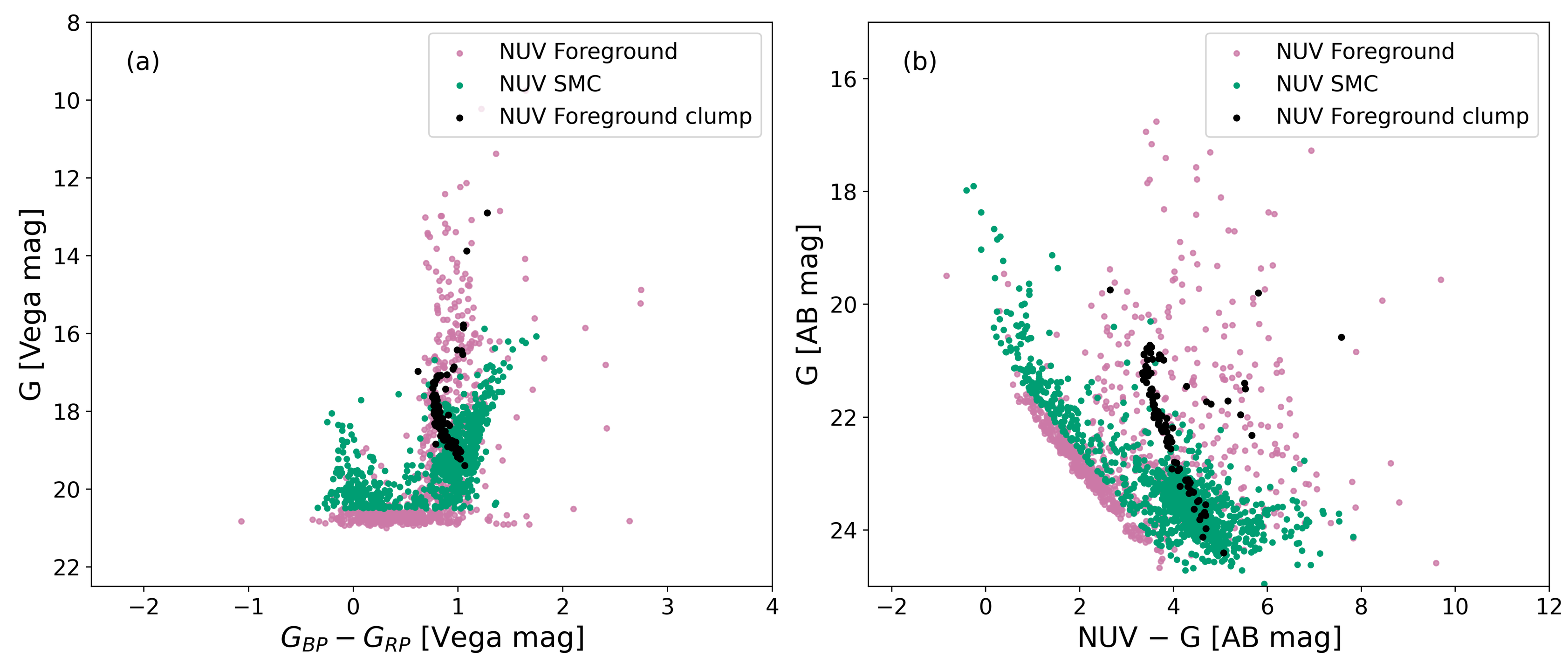

Figure 4. **CMDs of NUV sources:** (a) **Optical**, (b) **NUV–optical**. In both, black points indicate the stars in foreground clump within the ellipse shown in the Figure 2.

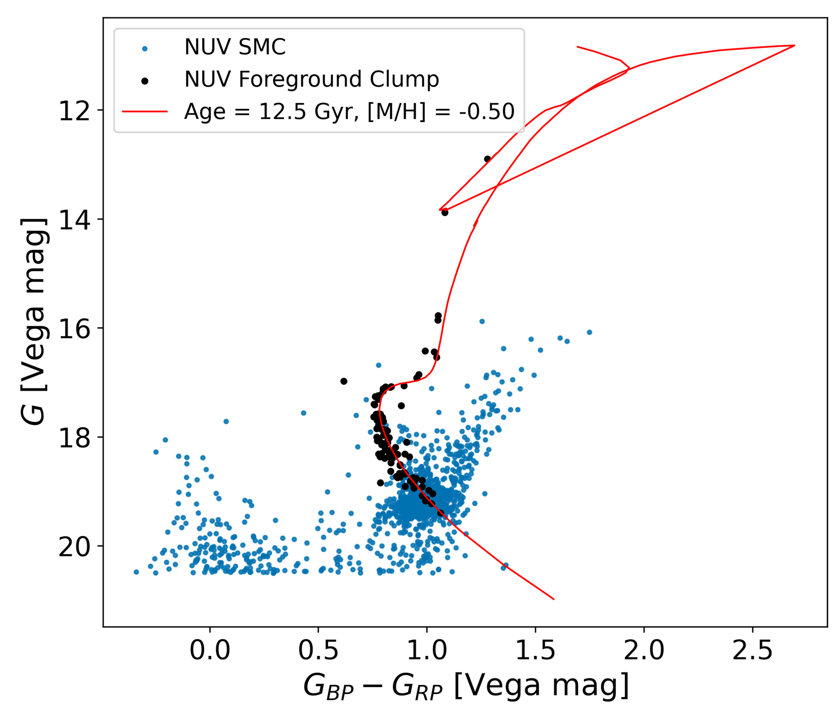

Figure 5. **Optical CMDs** of the **foreground clump** (Fig 2) in Black and **NUV-SMC** in Blue. Red lines show the 47 Tuc isochrone, matching the age and metallicity of the foreground clump, and the stars along this line have proper motions similar to those of 47 Tuc; together, these confirm the clump as 47 Tuc. The diagram also demonstrates how the 47 Tuc is passing through the red clump region of SMC.

Conclusions

- **Population Disentanglement:** UVIT and Gaia effectively separate the SMC, 47 Tuc and other foreground stars.
- **Kinematics:** FUV-Gaia detects SMC members and diffuse foreground population, whereas NUV-Gaia Reveals addition Foreground clump.
- **47 Tuc:** The foreground Clump is confirmed as 47 Tuc based on Proper motion and isochrone.
- **UV Sensitivity:** FUV detects mainly hot main sequence stars, while NUV also includes cooler stars.
- **Multi-wavelength Analysis:** Essential to isolate SMC members and study tidal history.

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